Tanzim Hossain Romel

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Uttara, Dhaka, Bangladesh

Research Interests

AI for Software Engineering (AI4SE), Software Engineering for AI (SE4AI), Trustworthy AI, LLM Agent Security, Empirical Software Engineering, Developer Tooling

Education

- **University of Alberta** — *Starting Sep 2026*
  Incoming M.Sc. in Computing Science — Edmonton, Canada
  - Incoming member of the U-A-Goose research group in the Department of Computing Science.
- **Bangladesh University of Engineering and Technology (BUET)** — *Mar 2018 - May 2023*
  B.Sc. in Computer Science and Engineering — Dhaka, Bangladesh
  - Thesis: Patient-Centric Blockchain Framework for Electronic Health Record Management

Work Experience

- **IQVIA** — *June 2023 - June 2026*
  Software Development Engineer — Dhaka, Bangladesh
  - Backend Engineer developing microservices-based healthcare applications handling millions of patient records using .NET Core, C#, and AWS
  - Deployed multi-agent systems using LangGraph, MCP, and RAG for dashboard generation/modification, integrated with data exploration agents achieving 85% reduction in setup time
  - Achieved 60% reduction in query execution times through database optimization; implemented 40% API response improvement via Redis caching
  - Contributed to browser automation testing methodology in .NET, simplifying regression testing and improving test coverage from 72% to 95%
  - Received IQVIA Impact Program – Silver award (May 2025) for outstanding performance

Research Experience (Selected)

- **An Empirical Study on Remote Code Execution in ML Model Hosting Ecosystems** — *June 2025 - Oct 2025*
  Tools: Python, Bandit, CodeQL, Semgrep, YARA, CWE Analysis | Submitted to TOSEM 2026
  - Studied remote-code-execution risk across ~45,000 ML model repositories with Mohammad Latif Siddiq and Joanna C. Santos.
  - Used Bandit, CodeQL, Semgrep, and YARA to find vulnerability patterns and explain 600+ related developer discussions.
- **The Choice Can Be the Attack: Auditing Aligned Backdoors in LLM Agents** — *August 2025 – Present*
  Tools: Python, vLLM, PyTorch, pandas, NumPy
  - Studied a hidden backdoor risk where a trigger changes which valid option an LLM agent chooses while the final answer still appears correct.
  - Built **SHIFT**, which reruns the same task with and without a known trigger, then checks whether the changed choice favors the attacker's target rather than normal reasons like price or rating.
- **Multi-Agent Framework for Generating Relational DB Schema & ERD** — *July 2025 - Present*
  Tools: Python, LangGraph, StateGraph, Z3 Solver, SQLAlchemy, Text2Schema
  - Extended SchemaAgent with Dr. Sukarna Barua (BUET) to turn natural-language requirements into relational schemas and ER diagrams.
  - Built a LangGraph pipeline for entity extraction, relationship detection, normalization checks, Z3 validation, and targeted retry.

Projects

- **ctxhelm: agent-native context compiler** *May 2026 – Present*
  Tools: Rust, MCP, CLI, Git, Hybrid Retrieval, Graph Signals, Empirical Evaluation
  - Designed and implemented a local-first context-retrieval engine for AI coding agents, combining safe repository inventory, lexical and symbol search, graph expansion, semantic-provider provenance, related-test mapping, git-history signals, and memory/feedback metadata.
  - Built an empirical evaluation and release-validation harness with fixed-budget retrieval studies, lexical baselines, signal ablations, source-free proof artifacts, deterministic MCP protocol checks, and real-client Claude Code evidence for `prepare_task` and `get_pack`.
- **Yet Another C Compiler** *Jun-Aug 2021, Oct 2024*
  Tools: C++17, CMake
  - Built a C compiler with lexer/parser, semantic analysis, x86-64 code generation, and a linear-scan register allocator.
  - Modernized the Flex/Bison prototype with modular IR passes and SSA optimizations including SCCP, GVN, LICM, and dead-code elimination.
- **Bangla Handwritten Digit Recognition** *Jan 2023 – Feb 2023*
  Tools: Python, NumPy, OpenCV, CNN from Scratch, NumtaDB
  - Implemented a NumPy CNN from scratch to recognize handwritten Bangla digits from the NumtaDB dataset.
  - Built the preprocessing and training pipeline, reaching **95.9% accuracy** and placing **2nd among 120 students**.
- **Eventfly: End-to-end Event Management System** *May 2022 - July 2022*
  Tools: TypeScript, Express.js, Next.js, Docker, Kubernetes, NATS, MongoDB
  - Designed microservices-based event management system
  - Led back-end architecture implementing newsfeed, payment, authentication, and event management services

Open Source Contributions

- **Selected Upstream Contributions** *2025 – 2026*
  Selected upstream contributions across SRE benchmarks, program analysis, .NET, web servers, agent systems, and compilers GitHub
  - **SREGym**: authored and hardened Kubernetes SRE benchmark work, including a scheduler priority-preemption cascade scenario for Hotel Reservation and a Calico route-reflector label-drift follow-up with RFC, fault/oracle design, validation checks, and reviewer-facing evidence.
  - **RefactoringMiner**: merged merge-parent-aware commit diffs, advanced assertion migration matching, and single-page PR viewed toggles; currently proposing a read-only local MCP server for refactoring analysis and intent validation.
  - **EF Core**: merged runtime migration creation/application, `ON DELETE SET DEFAULT` support, and nullable complex-property reload handling.
  - **GenHTTP**: added automatic request decompression and binary-response support.
  - **deepagents**: merged a CLI fix that stops and clears `LoadingWidget` animation timers on stop and unmount, preventing interval leaks in long-running terminal sessions and adding regression coverage for both cleanup paths.
  - **TypeScript**: merged an upstream-aligned declaration-emit alias-resolution fix that preserves imported type aliases for inferred exports across module boundaries in Microsoft's Go port of TypeScript.

Skills

- **Agentic AI & LLM Tooling:** MCP, LangGraph, LangChain, RAG, vLLM, tool calling, LLM evaluation, context engineering
- **Backend & Data Systems:** C#/.NET, REST APIs, PostgreSQL, MongoDB, SQL Server, Redis, SQLAlchemy, Oracle/PLSQL
- **Research & Security Tooling:** CodeQL, Semgrep, Bandit, YARA, Z3, RefactoringMiner, static analysis
- **Core Languages:** Rust, Python, C#, TypeScript, C++, Go, Java, SQL, Solidity
- **Cloud & Engineering Infrastructure:** AWS, Docker, Kubernetes, GitHub Actions, Terraform, OpenTelemetry, Jaeger, Linux, Git

Honors and Awards

- **IQVIA Impact Program – Silver Award**: outstanding performance and essential feature delivery *May 2025*
- **Finalist, Blockchain Olympiad Bangladesh**: top 40 national team for blockchain-based ticketing *2021*
- **2nd Place – Bangla Handwritten Digits Recognition**: 95.9% accuracy, BUET ML Lab contest *2022*
- **Dean's List Award**: academic performance, BUET Level 2
- **National Science Olympiads**: prize winner in Bangladesh Physics and Chemistry Olympiads *2017*
- **Talentpool HSC Scholarship**: 15th in Rajshahi Board with 95.6% marks *2017*

References

Dr. Chowdhury Md. Rakin Haider
Assistant Professor, CSE, BUET

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Dr. Zhou Yang
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